

Homework — 1.1.3 Input, Output and Storage Devices

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.1.3 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Complete the table: classify each storage device as **magnetic**, **flash** or **optical**, and give one reason for your choice based on how the device stores or reads data. [3] (AO1)

Device	Magnetic / Flash / Optical	Reason for your choice
Blu-ray disc	(a)	(b)
SSD inside a laptop	(c)	(d)
External hard disk drive	(e)	(f)

2. Complete the passage about optical storage using words from the word bank. Not all words are used. [2] (AO1)

Data on an optical disc is stored as a spiral track of (a) _____ and lands on the disc surface. To read the disc, a (b) _____ is shone at the surface and the amount of (c) _____ light is measured; the difference between the two surface features is interpreted as (d) _____ data (1s and 0s).

Word bank: binary, laser, magnetised, pits, reflected, sectors, transistors

3. Tick **True** or **False** for each statement. If a statement is false, write a correction in the final column. [3] (AO1/AO2)

Statement	True / False	Correction if false
RAM is non-volatile, so it keeps its contents when the power is turned off.		
ROM typically stores the firmware/boot program used to start the computer, and its contents cannot normally be changed.		
With virtual (cloud) storage, files can only be accessed from the computer on which they were originally saved.		

4. A museum needs to **archive** several terabytes of footage that will rarely be accessed again, at the **lowest possible cost**. Recommend a suitable storage medium and **justify** your choice against the requirements. [3] (AO2)

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5. A supermarket self-checkout must carry out the three tasks below. State a suitable device for each task.

(a) Read the barcode on each product. [1] (AO2)

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(b) Show the running total to the customer. [1] (AO2)

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(c) Produce a paper receipt. [1] (AO2)

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Part B — Retrieval: keep it fresh (~6 marks)

6. State **two** differences between **RISC** and **CISC** processors. [2] (1.1.2)

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7. Explain why a **GPU** is well suited to applying the same calculation to a large data set. [2] (1.1.2)

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8. State the **full name** and **role** of the **MDR**. [2] (1.1.1)

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